Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Safety Intelligence for Humanoid Robots			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials			
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	Robot Control and Intelligence Technologies		
Concept of Proposed Technology	The project will develop **Safety Intelligence, a novel technology that enables humanoid robots to autonomously understand, predict, and mitigate physical risks during operation in human environments**. Unlike traditional robot safety approaches, which rely on predefined rules, fenced workspaces, or fixed risk assessments, Safety Intelligence introduces a dynamic, AI-driven safety layer embedded directly into the humanoid's control architecture. This technology will combine: - AI-based predictive risk analysis that continuously estimates potential collision risks - Real-time biomechanical modeling to assess the physical consequences of contacts - Autonomous motion adaptation that keeps robot behavior within safe biomechanical limits - Model-based validation mechanisms ensuring compliance with international safety standards The result is a self-monitoring humanoid system capable of adjusting its motion, speed, and			

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	interaction behavior in real time to maintain safe operation, even in changing and unpredictable environments. Safety Intelligence therefore represents a shift from external safety enforcement to intrinsic, cognition-based safety control inside the robot.
Rationale for Joint R&D Project	Developing Safety Intelligence for humanoid robots requires the integration of advanced AI methods, biomechanical modeling, real-time control systems, and safety validation technologies. No single organization can cover this interdisciplinary scope alone. Fraunhofer IFF provides strong expertise in: - safety-critical system design - biomechanical modeling and collision analysis - compliance with international safety regulations and standards A Korean SME partner brings: - agile industrial development capabilities - strong safety and AI engineering expertise - fast technology transfer into robotic products South Korea has a globally recognized strength in rapid robotics commercialization and is highly receptive to next-generation intelligent automation technologies. Korean SMEs are often early adopters of AI-driven robotics innovations and can accelerate the transition of Safety Intelligence from lab to market. This bilateral collaboration ensures: - excellence transfer-ready research - rapid engineering and product integration - early access to European robotics markets by Korean companies
Key Words	1. Humanoid robots 2. Safety 3. Human-centered technology 4. AI

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The project aims to develop the following core technologies: **Predictive Risk Intelligence Engine** AI-supported system that forecasts potential collision scenarios based on robot motion and environmental dynamics. **Digital Biomechanical Collision Model** Real-time model to estimate impact forces and ensure they remain below safety thresholds. **Autonomous Safe Motion Generator** Control module that continuously adapts humanoid motion parameters to maintain safe interaction limits. **Integrated Safety Intelligence Architecture** Unified software framework embedding risk prediction and mitigation directly into humanoid control loops. **Validation & Safety Assurance Toolkit** Digital tools to verify compliance with international collaborative robot safety standards.			
R&D Target Specifications	Risk prediction horizon Motion adaptation latency Biomechanical force estimation error Safe speed regulation accuracy Real-time processing capability Technology Readiness Level	≥ 500 ms ahead of potential collision ≤ 50 ms response time ≤ 10% deviation from measured values Maintain forces within ISO/TS 15066 limits with ≥ 95% reliability ≥ 100 Hz safety evaluation cycle TRL 6 (validated in relevant humanoid scenario)		
R&D Plan	PLEASE NOTE: The following RnD plan is a draft that has to be detailed with the Korean SME later. **WP1 Project Management** (M01 to M36) - Coordination and technical progress monitoring - Risk, quality, and IP management - Reporting and dissemination **WP2 Requirements & System Design** (M01 to M24) - Definition of functional and safety requirements - Development of Safety Intelligence system architecture - Design of integration and validation methodology **WP3 Simplified Risk Analysis** (M01 to M24) - Collection and formalization of collision knowledge - Development and training of AI-based risk prediction - Simulation-based validation of predictive performance **WP4 Automated Risk Reduction and Validation** (M12 to M24) - Development of real-time safe motion generator - Creation of dynamic humanoid collision model - Experimental validation of force and speed regulation **WP5 Validation** (M24 to M36) - Integration into a humanoid robot platform - System-level testing in dynamic environments - Final TRL 6 demonstration and optimization			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	PLEASE NOTE: The following RnR are based on assumptions that have to be clarified once the project is detailed with a Korean SME. Fraunhofer IFF will: - Lead biomechanical modeling and collision analysis - Develop and validate safe motion control algorithms - Ensure compliance with European safety standards and perform system validation - Provide expertise in AI and humanoid robots - Scientific excellence Resources provided (negotiable): - Background IP (technologies, patents, etc.) - Laboratories for experimental validation - Researchers from various domains
Expected R&R of Korean SME Partner	PLEASE NOTE: The following RnR are based on assumptions that have to be clarified once the project is detailed with a Korean SME. The Korean SME will: - Develop AI-based risk prediction and software components - Implement real-time integration into humanoid control systems - Lead system integration, testing, and commercialization preparation - Use Case and Impact Resources provided (negotiable): - Use case - Humanoid robot for testing - Robotics engineers

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	PLEASE NOTE: The following outcomes do not consider any interest from the partner SME. **Technological Outcomes** - First integrated Safety Intelligence framework for humanoid robots - Real-time AI-driven intrinsic safety control - TRL 6 validated collision prediction and mitigation system **Commercial Impact** - Reduced safety engineering effort for humanoid deployment - Faster certification and market readiness - Strong commercialization pathway via Korean SME **Industrial & Economic Impact** - Strengthened German-Korean leadership in safe intelligent robotics - Accelerated adoption of humanoid robots in service and industry - Contribution to emerging global markets for autonomous humanoid systems

6. References

The following references are considered relevant for the work:

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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